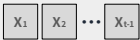
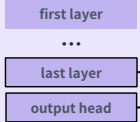


Input Tokens



Base Model f_B



Always run

1

Hierarchical Router

Macro Encoder
single pass

concatenate

Micro Router
token level

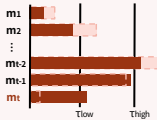


$$p_t = \sigma(\text{MLP}_{\text{micro}}([h_t^B; \text{MLP}_{\text{macro}}(c)]))$$



2

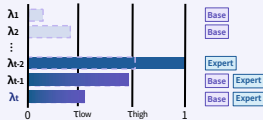
Momentum Smoothing



$$m_t = \alpha m_{t-1} + (1 - \alpha) p_t$$

3

Threshold-gated Dispatch



$$\lambda_t = \begin{cases} 0 & \text{if } m_t < \tau_{\text{low}} \\ m_t & \text{if } \tau_{\text{low}} \leq m_t \leq \tau_{\text{high}} \\ 1 & \text{if } m_t > \tau_{\text{high}} \end{cases}$$

Output token x_t

$$p(x_t | x_{<t}) = (1 - \lambda_t) p_b + \lambda_t p_e$$

Expert Model f_E

Invoked only on critical points

: Frozen

: Trainable